

Pre-design solution: Тестовый отказ системы

Region: Новосибирская область **Greenhouse type:** year-round **Crop:** tomato **Target yield:** 2000.0 t/year **Generated:** 2026-06-17 06:36

1. Site inputs and analysis

Target yield per plot area: 4000.0 kg/m²/year $\triangle$ Target yield is very high — intensive technology required.

Climate parameters (per Russian SP 131.13330):

Parameter	Value
Design winter temperature (t5)	-37.0 °C
Design summer temperature	25.0 °C
Heating degree-days	6076.0
Heating period	227 days
Snow load	2.4 kPa
Wind load	0.3 kPa
Solar radiation (winter)	1.5 MJ/m ² /day

2. Design solution (variant failed_v1)

Rationale: This is a year-round glass greenhouse complex designed for intensive tomato production (trellis culture) in the severe continental climate of the Novosibirsk region. Glass cladding is selected to maximise light transmittance ($\tau \approx 0.88$), which is critical given the low winter solar radiation. The brief explicitly requires at least two separate blocks; however, the plot is extremely small (25 × 20 m) and must accommodate year-round greenhouses with a mandatory minimum 6 m clearance between blocks per SP 107.13330 clause 4.4. Two blocks are arranged along the 25 m plot length with a 6 m fire-and-access gap between them, leaving each block with the minimum viable footprint. Important engineering caveat: the clay soil with groundwater depth of only 0.5 m (frost-heaving conditions) requires a pile foundation with perimeter drainage and enhanced waterproofing. It must be honestly noted that the 500 m² plot is physically incapable of accommodating the target yield of 2 000 t/year — even with the most intensive technology, the available growing area yields a small fraction of that figure; this constraint (ENG.5) cannot be resolved by design choices alone and must be escalated to the client for site re-selection or yield target revision.

Total growing area: 216 m² **Complex footprint (with auxiliary):** 500.0 m²

Greenhouse blocks

- **Block A** — 12.0 × 9.0 m (area 108 m²)
- Heights: eave 2.4 m, ridge 5.5 m
- Layout: block (multi-span), 1 spans of 9.0 m
- Roof: straight, slope 45.0%
- Covering: glass ($\tau=0.88$), glass 4.0 mm - Opaque structures share: 14.0%
- **Block B** — 12.0 × 9.0 m (area 108 m²)
- Heights: eave 2.4 m, ridge 5.5 m
- Layout: block (multi-span), 1 spans of 9.0 m
- Roof: straight, slope 45.0%
- Covering: glass ($\tau=0.88$), glass 4.0 mm - Opaque structures share: 14.0%

Complex structural parameters

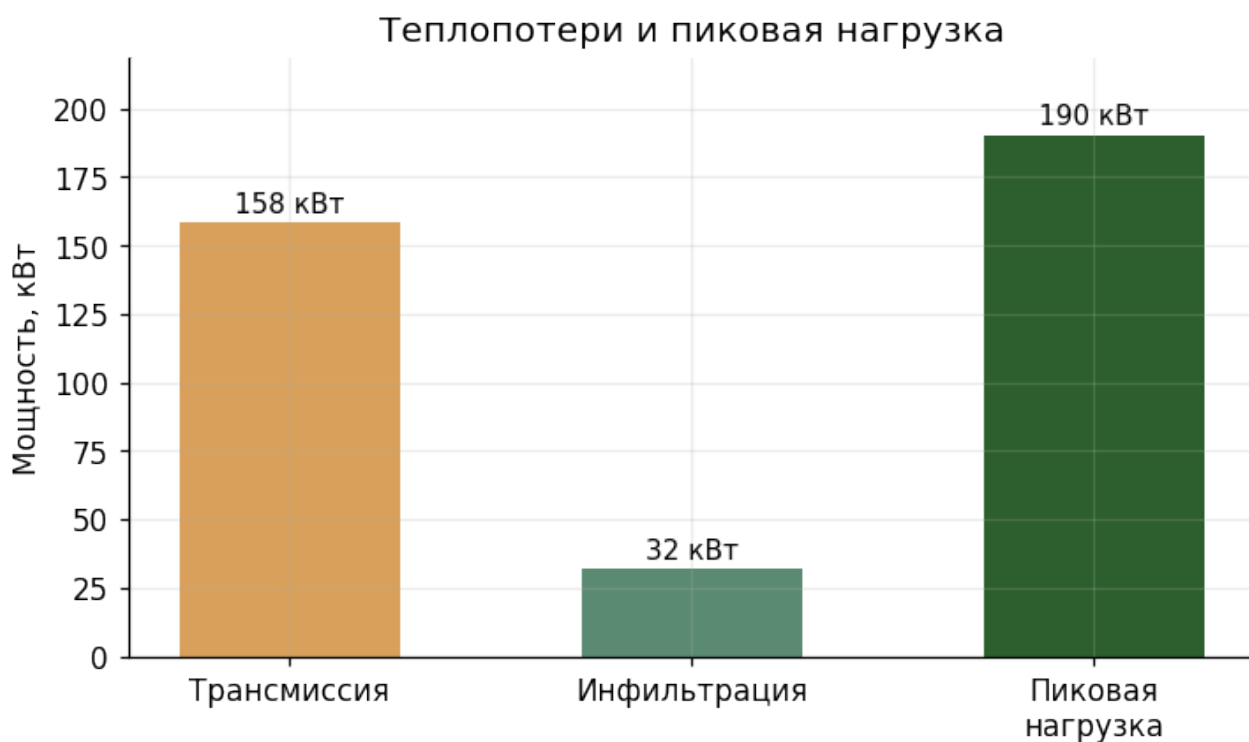
Parameter	Value	SP Clause
Plinth height	0.4 m	5.8
Foundation rise above soil	0.4 m	5.9
Territory fence height	1.8 m	4.16

Auxiliary zones

- **Boiler house** — 16.0 m² (Heating plant and utility room)
 - **Storage** — 12.0 m² (Storage for tools, fertilisers and consumables)
 - **Staff facilities** — 8.0 m² (Changing rooms and sanitary facilities for personnel)
 - **Nutrient mixing room** — 9.0 m² (Nutrient solution preparation and dosing pumps)
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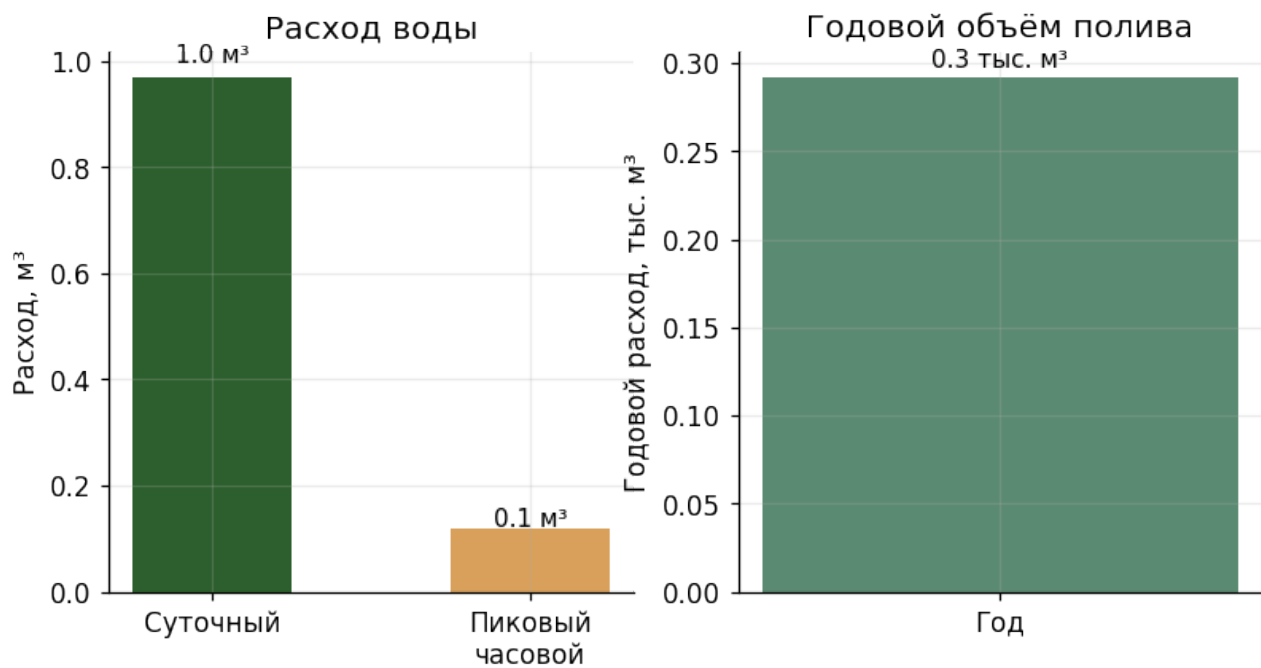
3. Engineering calculations

3.1. Heat supply



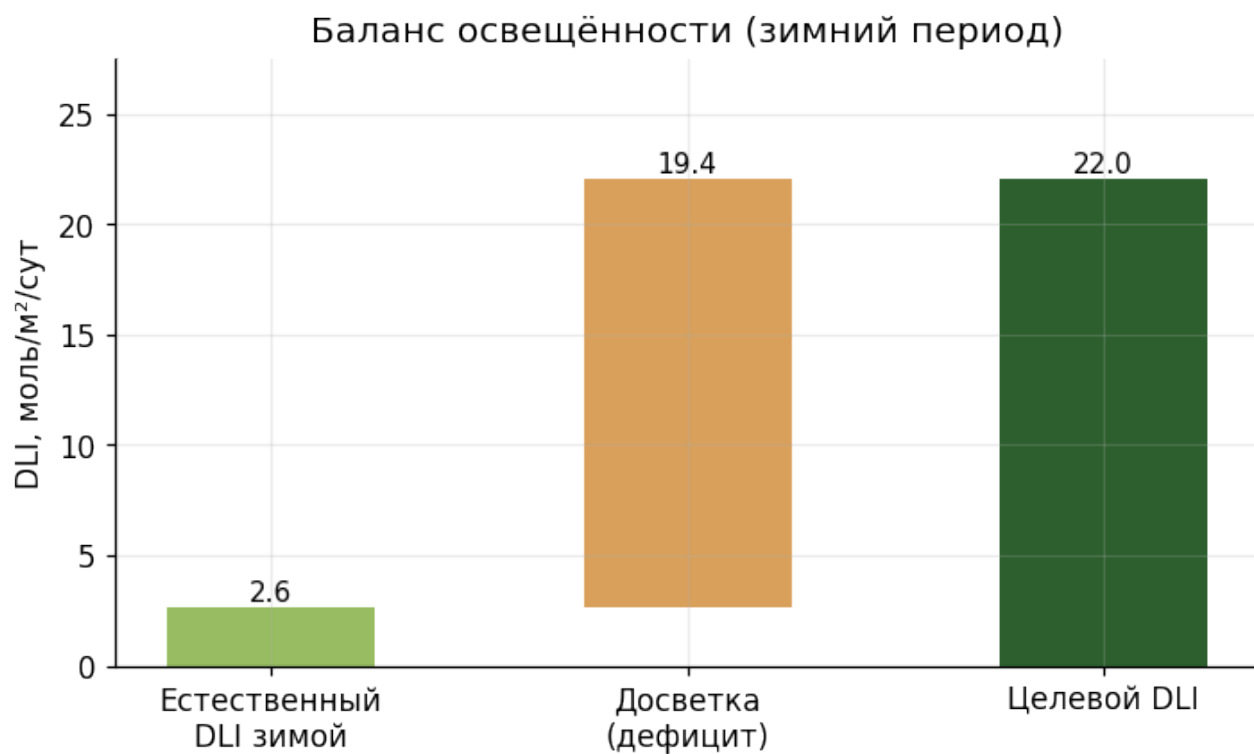
- Design temperature difference: **55.0 °C**
- Envelope area: **450.0 m²**
- Overall heat transfer coefficient U: **6.4 W/(m²·K)**
- Transmission losses: **158.4 kW**
- Infiltration losses: **31.7 kW**
- **Peak heating load: 190.1 kW**
- Annual heat demand: **504.0 MWh**
- Coolant temperature: **95.0 °C** (SP 7.9 ≤150)
- Lower-zone heat share: **45.0%** (SP 7.13 ≥40)

3.2. Water supply



- Daily demand: **0.97 m³/day**
- Peak hourly: **0.12 m³/h**
- Annual: **292 m³/year**
- Irrigation method: Drip (default)
- Reliability category: **II** (SP 6.14)
- Hose service radius: **40.0 m** (SP 6.8 ≤45)

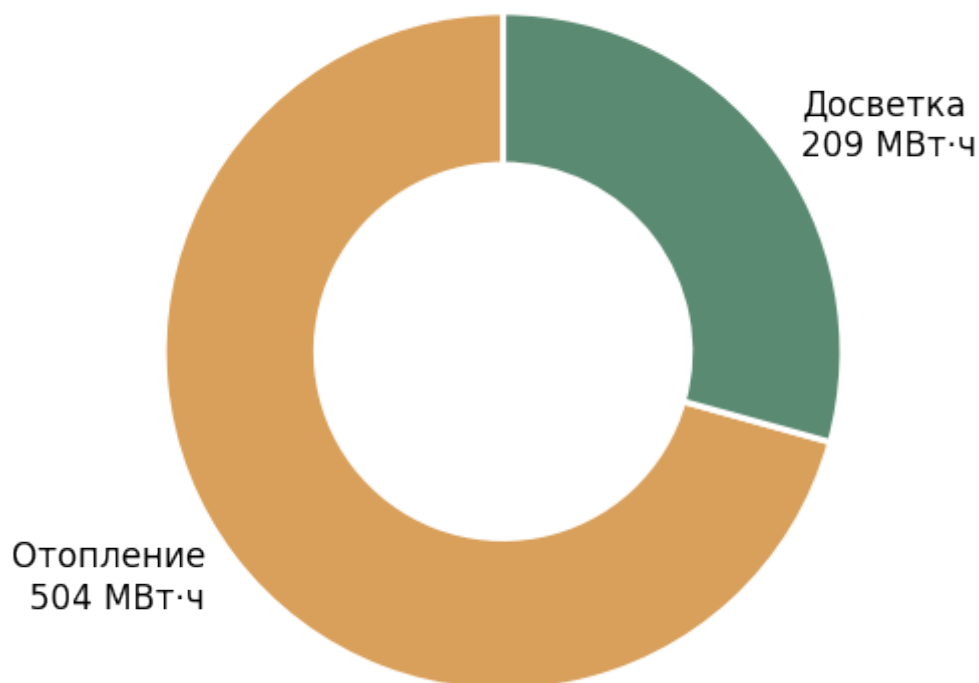
3.3. Lighting and supplemental



- Target DLI: **22.0 mol/m²/day**
- Natural winter DLI: **2.6 mol/m²/day**
- **Supplemental lighting required:** installed 266.2 W/m², consumption 208837 kWh/year
- Aisle floor illuminance: 8.0 lx (SP 8.3 ≤10)

3.4. Annual energy consumption

Годовое энергопотребление: 713 МВт·ч



3.5. Ventilation

- Summer target air changes per hour: **60.0 h⁻¹**
- Vent opening share of envelope: **20.0%** (SP 7.18 ≥ 20 , ≥ 10 north of 60°N)
- Vent opening share of floor: 41.7%
- Forced ventilation: **not required**

3.6. Structural loads

- Snow load on roof: **835.0 kN** ($\gamma=1.4$)
- Wind load on walls: **48.0 kN** ($q_{10}=1.0$, $q_2=0.6$)
- Trellis load: 150.0 N/m² ($\gamma=1.3$)

Overload factors and normative values per SP 107.13330 clause 5.14.

3.7. Geotechnical conditions and foundation

- Foundation soil: **clay**
- Groundwater depth: **0.5 m**
- Recommended foundation type: **pile**
- Minimum embedment depth: **3.0 m**

- Perimeter drainage: **required**
- Waterproofing grade: **enhanced**

Clay with high groundwater table — frost-heaving soil. Pile foundation embedded below the freezing depth; enhanced waterproofing and mandatory perimeter drainage required.

Preliminary recommendations. A proper engineering-geological survey per SP 47.13330 is required for working documentation.

4. SP 107.13330 compliance check

Rules checked: **36**

ERROR — ENG.5-yield-feasibility

Площадь выращивания × норма культуры должна покрывать минимум 90% целевой урожайности из ТЗ. Иначе участок физически не вмещает заявленный объём продукции.

- Actual: вмещает ~11 т/год (tomato @ 50.0 кг/м² × 216 м²)
- Required: >= 1800 т/год (90% от ТЗ 2000.0 т)
- **Source** (SP 107.13330, original Russian text): Инженерная проверка (не из СП): достижимость целевой урожайности

Generated by agro-greenhouse-designer. Outputs are pre-design estimates and require validation by a qualified engineer before detailed documentation.